

Virtual Dai Realm: AI Mocap and XR Staging for Culturally-Situated Embodied Performance

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Figure 1: Conceptual Rendering of Virtual Dai Realm.

ABSTRACT

Virtual Dai Realm is an immersive VR artwork that reimagines Dai ethnic dance—an intangible cultural heritage grounded in animistic beliefs and nature-mimicking movement—through AI motion capture and XR staging. Motion data extracted from a professional dancer drives responsive wind fields, particle blooms, cloth dynamics, and light behavior, while audience proximity and gaze subtly shape the unfolding scene. Viewers shift from spectators to co-ritualists, completing the work through embodied participation. By bridging ancestral motion (past), lived bodily encounter (present), and AI-mediated authorship (future), the piece demonstrates how immersive technologies can foster emotionally grounded connections to cultural heritage.

Index Terms: Virtual Reality; AI Motion Capture; Immersive Dance; Intangible Cultural Heritage; Gesture-Based Interaction.

1 DESCRIPTION OF ARTWORK

Virtual Dai Realm is an immersive VR artwork that reimagines the spiritual poetics of Dai dance through AI motion capture and XR virtual production. The Dai are a cross-border ethnic group in Southwest China and the Mekong region, whose aesthetics are tightly bound to rivers, monsoon forests, and seasonal rituals. Dai dance often mimics animals and natural forces, embodying a worldview of *wanwu youling* (“all things have spirit”). In our work, these motifs are translated into an embodied, multisensory environment where gesture, airflow, light, and particulate matter interweave as one living choreography.

Structurally, the piece unfolds as four interlinked acts—Wind, Flower, Mirror, and Moon—each mapping a traditional image to a contemporary interaction grammar [6]. *Wind* stages vitality and freedom through responsive gust fields; *Flower* celebrates renewal

via blooming particle systems; *Mirror* becomes a site of self-encounter through a stylised digital twin; *Moon* resolves the journey in nocturnal stillness with cloth, fog, and firefly swarms. These scenes draw directly from the project’s creative dossier and production pipeline: AI mocap of a professional performer (DeepMotion), green-screen video capture, camera solving, Niagara particles for blossoms and snow, Style3D cloth / soft-body simulation, TouchDesigner audio-visual mapping, and Unreal Engine real-time compositing for VR.



Figure 2: Human dancer interacting with digital avatar.

Rather than presenting heritage as a static archive, *Virtual Dai Realm* treats ritual as relational experience. The artwork asks: What if ancestral motion could breathe again through computational choreography? Participants enter a spirit-garden where their presence shapes the environment. By stepping closer, they strengthen the wind field; by sustaining their gaze, they stabilise the mirrored avatar’s focus (as shown in Fig. 2); by drifting back, they dissolve the scene into moonlit mist. The piece transforms spectatorship into co-performance: audience bodies complete the rite, and in doing so, rediscover the emotional bonds that connect culture, landscape, and memory.

This thematic orientation resonates with Dai cosmology: nature is animate; water purifies; cycles of season and self are continuous. Our dramaturgy draws on Dai “water culture” (e.g., the cleansing symbolism at the core of major festivals) and the long-standing figuration of spirits of wind, trees, and flowers. These images inform both digital character design (e.g., spirit personae like the “snow spirit”) and environmental behaviors (e.g., feather-snow and red-silk wind cues), allowing viewers to sense a continuum between human movement and non-human agency.

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2 EMOTIONAL CONNECTION THROUGH IMMERSION

Virtual Dai Realm treats immersion as a bridge from recognition to participation. Instead of explaining Dai imagery, the work makes it felt: you sense wind pressure in the soundfield; you inhabit the flower's bloom as particulate swirls around your arms; you confront your mirrored twin as an uncanny, stylised self; you breathe into the calm of moonlight. This progression is designed to move audiences from witnessing heritage to animating it, echoing the Dai belief that spirit is not owned but co-present [3]. The result is empathy not through exposition, but through shared motor intention—a sustained, embodied relation between past gestures and present bodies.

Moreover, the project positions AI mocap as a cultural interlocutor rather than a mere production tool. By capturing and stylising dance kinematics, AI helps isolate gesture primitives—curves of the wrist, arcs of the spine, suspensions and drops—that carry cultural semantics. In VR, these primitives drive wind vectors, particle births, cloth tension, and light scattering in real time [5]. Thus, historical motion becomes a procedural score that not only preserves but extends meaning: audiences can re-compose the work through proximity, gaze, and pacing, generating new yet culturally anchored variants of the rite. The piece proposes a future where immersive systems act as vessels of continuity—not freezing tradition but letting it evolve through co-creation with visitors and machines [1].

The work avoids nostalgia and ethnographic literalism in favor of embodied continuity [4]. The emphasis is on affect, coherence, and repeatability, demonstrating how ancestral motion data can orchestrate contemporary sensation, and how viewer agency can sustain a living ritual ecology.

3 IMPLEMENTATION AND TECHNICAL EXECUTION

3.1 Capture and Stylisation

Green-screen video of the dancer; AI motion capture (DeepMotion) to extract full-body kinematics [2]; cleanup and stylisation in DCC tools; import into UE5 (As shown in Fig. 3).



Figure 3: Workflow for integrating green-screen footage into virtual environments.

3.2 XR Staging

Real-time compositing plus scene layers for the four acts. Niagara powers blossoms, snow/feather morphs, fireflies, and mist; Style3D cloth / soft-body simulates red-silk and garments; TouchDesigner drives audio-reactive parameters and light cues. A point-cloud aesthetic is used as a transitional “medium” that sutures live-action silhouettes and digital spirits, smoothing appearance/disappearance and guiding attention.

3.3 Interaction Model

Proximity increases wind intensity and particle birthrate in Wind; hand presence and head stability focus the mirrored twin in Mirror;

radial drift modulates bloom density and chroma in Flower; breath-paced stillness calms fog turbulence in Moon. These mappings are bounded and repeatable, favoring graceful response over game-like freedom to maintain dramaturgic clarity and comfort.

3.4 Performance and Comfort

Target 90 Hz with single-pass stereo; aggressive LODs on particle and cloth systems; dynamic resolution scaling on GPU spikes; audio spatialisation relies on the headset (no external speakers) to streamline setup and reduce bleed. Session length and locomotion are tuned to minimize discomfort while retaining agency.

4 ARTIST INFORMATION

Tianxing Zhou is a PhD Candidate at the School of Communication and Arts, The University of Queensland. Her research tackles the narrative paradox in VR interaction, developing approaches where gameplay and storytelling co-amplify rather than compete. She serves as a reviewer for ACM Multimedia 2025 and a Program Committee Member for AICS 2025. Her award-winning VR work *So Long* has been exhibited at over twenty venues worldwide. Her work also received the Best Sound Design Award at the UNESCO Digital Heritage Contest. She holds a Master's degree in Immersive Storytelling from University College London, where she received the Best Dissertation Award and was named to the Dean's List.

Zihan Chen is a graduate student at the School of Intelligent Image Engineering, Beijing Film Academy. His primary research focuses on AI-enabled optimization throughout the film and television production pipeline, virtual production, and XR-based shooting. His works have been shortlisted for a host of renowned awards, including selections for the Milan Design Week, nominations for the UK Culture and Arts Awards, and finalist positions at the National Digital Design Competition of China.

Xiaoxuan Liu is a graduate student at the Beijing Film Academy Digital Media School, specializing in Game Creation. During her studies, she was awarded the National Scholarship and recognized as an Outstanding Graduate of Beijing, among other honors. Her artistic practice focuses on VR digital art, and she has established a workflow centered on interactive engine technology. Her works have garnered dozens of awards and nominations, including the Outstanding Award at the 32nd College Student Film Festival of the Beijing International Film Festival and the Gold Award at the Busan International Art Festival.

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